Hyper-parameter	Description	Impact
Learning Rate	Size of the update or step	• It affects how quickly the model learns. • Steps that are too big can make learning unstable. • Steps that are too small can make learning too slow.
Batch Size	Number of examples the model looks at before making an update	• It affects how smooth the learning is and how much memory is used • Bigger batches make learning smoother but use more memory. • Smaller batches may reduce the quality of the model.
Number of Epochs	Number of times the model looks at the entire dataset	• It affects how well the model learns • More epochs can improve learning but might lead to overfitting, where the model performs well on training data but poorly on new data.

8.3.5 Fine-tuning Process

Finally, you are ready to fine-tune the model, and this would involve the following:

Initialise the Pre-Trained Model

Load and adapt the pre-trained model to your specific dataset.

Begin Training

Start training the model with your data. Monitor performance metrics like loss (how well predictions match outcomes) and accuracy (the percentage of correct predictions).

Adjust Hyperparameters

Modify settings such as learning rate, batch size, and number of epochs based on performance to improve results.

Iterate and Refine

Adjust and retrain the model as needed, evaluating its performance regularly to ensure effectiveness.